

Lab 3: Architectural Pattern Selection — Written Justification

Problem Statement #28 — Inter-City Freight Load Matching Marketplace

Architecture Selection: We chose the Microservices Architecture for the Inter-City Freight Load Matching Marketplace System.

Reason 1 — Independent Scaling for Uneven, Bursty Load

The marketplace has workloads with very different traffic patterns: bidding spikes sharply during each load's fixed auction window (FR-001), while payment settlement and proof-of-delivery processing occur steadily as shipments complete. A Microservices split lets the Bidding Service scale out independently during auction surges without over-provisioning the Payment & Escrow Service, which a monolithic Layered or centralized Client-Server design would force to scale together.

Reason 2 — Fault Isolation and Independent Deployment

The system spans genuinely separate concerns — load posting, bidding, carrier permit/insurance verification, delivery confirmation, and financial escrow — each with its own compliance and release cadence. Microservices let the Carrier Verification Service or Payment & Escrow Service be updated, audited, or temporarily degraded (e.g., a regulatory database outage) without taking down load posting or bidding, which is a single point of failure risk in a Client-Server design.

Security Advantage

Isolating the Payment & Escrow Service as its own microservice, with its own data store and stricter access controls, limits the blast radius if another service such as Bidding is compromised. Sensitive financial and escrow data never needs to be reachable from the same process or database as public-facing bid submissions, directly supporting the compliance intent behind NFR-002 (carrier permit and insurance verification).

Performance Benefit

Because the Payment & Escrow Service runs and scales independently of the Bidding and Load Posting Services, a surge in bid submissions during an auction window does not compete for the same compute or database resources as payment processing. This helps the system reliably meet the 60-second milestone payment disbursement target under peak load, as required by NFR-001.